

7. Noncompliance and Instrumental Variables

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Outline

1. Randomized Experiments with Noncompliance
2. Compliance Types
3. Instrumental Variables
4. Two-Sided Noncompliance
5. In-Class Exercise

Where are we? Where are we going?

- We've covered randomized experiments (no confounding)
- We've covered selection on observables (no unmeasured confounding)
- What if there is unmeasured confounding? What can we do?

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- We've covered randomized experiments (no confounding)
- We've covered selection on observables (no unmeasured confounding)
- What if there is unmeasured confounding? What can we do?
- First approach we will explore: **instrumental variables**
 - First: motivate IV through experiments and noncompliance.
 - Next: discuss how it relates to classical econometric methods like 2SLS?

Researcher: You, are in the control group. No need to take the treatment

Defier: But I want it!

Researcher: Just kidding, you are in the treatment group. Here it is

Defier:



Source: [Causal Inference for the Brave and True](#) By Matheus Facure Alves

1/ Randomized Experiments with Noncompliance

Noncompliance

- Get-out-to-vote (GOTV) experiment with door-to-door canvassing:



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- Households are randomized so treatment assignment is unconfounded
 - $Z_i = 1$ for assigned to treatment (canvassing attempted),
 - $Z_i = 0$ for assigned to control (no canvassing attempted).
- **Noncompliance:** units don't follow treatment assignment.
 - Units assigned to treatment take control or vice versa.
 - $D_i = 1$ for actually took treatment (heard canvasser message).
 - $D_i = 0$ for actually took control (didn't answer the door).
 - Full compliance means $Z_i = D_i$ for all i .

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- What do we do if we want the effect of *canvassing exposure* (D_i)?
- Two approaches one might consider:
 1. **Intent-to-treat** (ITT) analysis: just use randomization.
 - Use Z_i as the treatment and analyze as a typical experiment.
 - Downside: can't learn about the effect of actually being canvassed!
 2. **As-treated** analysis: just use treatment uptake.
 - As if D_i was randomly assigned.
 - Not valid if uptake is **correlated** with outcome.
 - $\rightsquigarrow$ unmeasured confounding between D_i and **POs**

The As-Treated Problem, Visualized

$n = 200$, $n_1 = n_0 = 100$, $\pi_{co} = 0.7$ (one-sided noncompliance)

$Z = 1$ group:

1	1	1	1	1	1	1	1
1	1	1	0	0	0	0	0

...

1	$D_i = 1$
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$Z = 0$ group:

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“Treated” ($D = 1$)

70 units

a selected subgroup

VS.

“Control” ($D = 0$)

130 units

everyone else

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↪ Alternative: leverage latent strata of **compliance types**.

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- Canvassing is an example of **one-sided noncompliance**.
 - People might refuse treatment when offered (i.e., $D_i(1) = 0$)
 - But no one receives treatment if in control (i.e., $D_i(0) = 0$, $\forall i$)
 - **Two-sided noncompliance** is when you can refuse to comply with treatment **or** control.

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 - $Y_i(1, D_i(1))$: potential outcome when assigned canvassing and whether uptake occurs for unit i when assigned to canvassing.
 - $Y_i(1, 1 - D_i(1))$: not possible to ever observe! (cross-world counterfactual)
 - Contradicts the unit's own compliance type.
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- Consistency assumption: $Y_i = Y_i(Z_i, D_i(Z_i))$

Some Notations

- Let's use o/1 subscripts for assignment, and t/c subscripts for uptake:

$$n_1 = \sum_{i=1}^n Z_i \quad n_0 = \sum_{i=1}^n (1 - Z_i) \quad n_t = \sum_{i=1}^n D_i \quad n_c = \sum_{i=1}^n (1 - D_i)$$

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$$\bar{Y}_1 = \frac{1}{n_1} \sum_{i=1}^n Z_i Y_i \quad \bar{Y}_0 = \frac{1}{n_0} \sum_{i=1}^n (1 - Z_i) Y_i$$

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- Assumption 1: **randomization** $[\{Y_i(d, z), \forall d, z\}, D_i(1), D_i(0)] \perp\!\!\!\perp Z_i$
 - For observational uses of IV, might condition on some $\mathbf{X}_i$.

Defining ITT Effects

- **Intent-to-treat** (ITT) effects are just the ATEs of Z_i :

$$\text{ITT}_D = \frac{1}{n} \sum_{i=1}^n D_i(1) - D_i(0) \quad \text{ITT}_Y = \frac{1}{n} \sum_{i=1}^n Y_i(1, D_i(1)) - Y_i(0, D_i(0))$$

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- Under randomization of Z_i , everything just like Neyman approach.
 - Variances, tests, CIs all standard.
- Problem: ITT_Y is a combination of true effect of D_i and noncompliance
 - Effect of D_i may be more **externally valid** than Z_i .

2/ Compliance Types

Compliance Status

- We can stratify units by their **compliance type**.
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 - Everyone has $D_i(0) = 0$, so no noncompliance there.
- Compliance is a function of potential outcomes so it is **pretreatment!**
 - $\rightsquigarrow$ treatment assignment independent of C_i

ITTs Among the Compliance Groups

- Compliance type indicator, $C_i \in \{\text{co}, \text{nc}\}$:
 - Number of compliers: $n_{\text{co}} = \sum_{i=1} \mathbf{1}(C_i = \text{co})$.
 - Proportion of compliers: $\pi_{\text{co}} = n_{\text{co}}/n$.
 - Same for noncompliers: n_{nc} and π_{nc} .

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- ITT on uptake is directly related to compliance type:

$$\text{ITT}_{D,\text{nc}} = \frac{1}{n_{\text{nc}}} \sum_{i:C_i=\text{nc}} D_i(1) - D_i(0) = 0$$

$$\text{ITT}_{D,\text{co}} = \frac{1}{n_{\text{co}}} \sum_{i:C_i=\text{co}} D_i(1) - D_i(0) = 1$$

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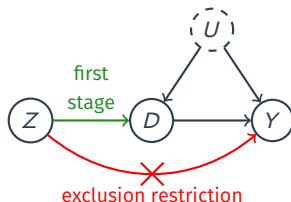
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- Implies that overall ITT on uptake equals the **proportion of compliers**:

$$\text{ITT}_D = \pi_{\text{co}} \cdot \text{ITT}_{D,\text{co}} + \pi_{\text{nc}} \cdot \text{ITT}_{D,\text{nc}} = \pi_{\text{co}}$$

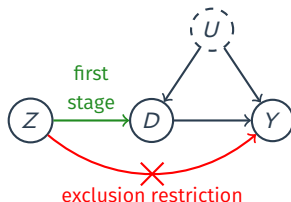
3/ Instrumental Variables

Identification Assumptions



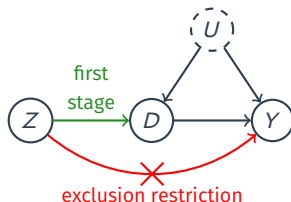
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- Implies that potential outcomes only a function of D_i :

$$Y_i(1) = Y_i(D_i = 1) = Y_i(Z_i = 1, D_i = 1)$$

$$Y_i(0) = Y_i(D_i = 0) = Y_i(Z_i = 1, D_i = 0)$$

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 - Low draft number could cause stress, change college plans, or alter career expectations, even before serving.
- ER is often the **most controversial** IV assumption.
 - Think carefully: is there *any* channel from Z to Y besides D ?

Outcome ITTs and Compliance Types

- We can define the ITTs on the outcome by compliance type as well.
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 - Implies that $ITT_{Y,co}$ is due entirely to treatment uptake.
- Allows us to connect the ITT_Y on the outcome to compliance groups:

$$\begin{aligned} ITT_Y &= \pi_{co} \cdot ITT_{Y,co} + \pi_{nc} \cdot ITT_{Y,nc} \\ &= ITT_D \cdot ITT_{Y,co} \end{aligned}$$

- Under the exclusion restriction, $ITT_{Y,co}$ is the effect of treatment receipt/uptake:

$$\begin{aligned} ITT_{Y,co} &= \frac{1}{n_{co}} \sum_{i:C_i=co} Y_i(1, D_i(1)) - Y_i(0, D_i(0)) \\ &= \frac{1}{n_{co}} \sum_{i:C_i=co} Y_i(D_i = 1) - Y_i(D_i = 0) = \tau_{LATE} \end{aligned}$$

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- This quantity is known as the **local ATE** (LATE), local to compliers.
 - It's a conditional ATE, where we condition on units being a complier.
 - Also referred to as **complier average causal effect** (CACE).

LATE

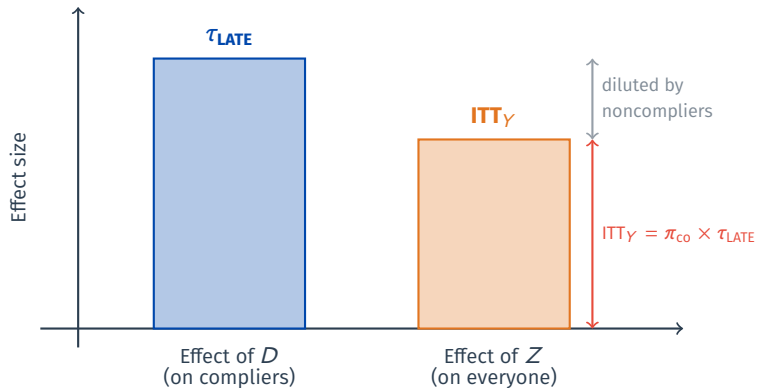
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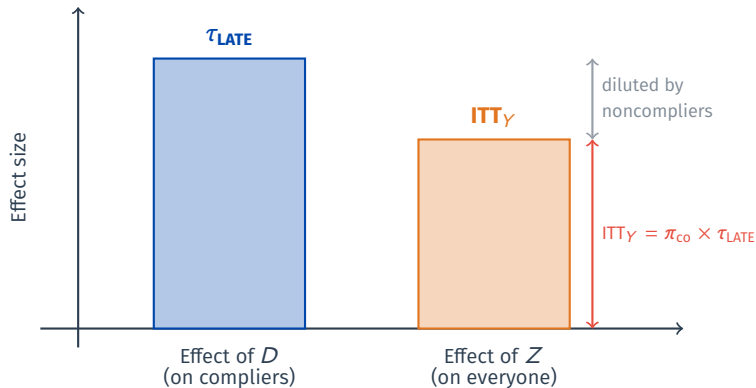
- This quantity is known as the **local ATE** (LATE), local to compliers.
 - It's a conditional ATE, where we condition on units being a complier.
 - Also referred to as **complier average causal effect** (CACE).
- LATE theorem** under one-sided noncompliance, exclusion restriction, first-stage, and randomization:

$$\tau_{LATE} = ITT_{Y,co} = \frac{ITT_Y}{ITT_D}$$

Noncompliance Dilutes the ITT



Noncompliance Dilutes the ITT



$\tau_{LATE} = ITT_Y / \pi_{co}$: the Wald estimator
"undoes" the dilution by dividing by π_{co}

- $\tau_{LATE} \geq ITT_Y$ always. Equal only when $\pi_{co} = 1$ (full compliance).

Wald Estimator

- **Wald** or **instrumental variable estimator** for the LATE:

$$\widehat{\tau}_{iv} = \frac{\widehat{ITT}_Y}{\widehat{ITT}_D}$$

- Ratio of the two unbiased ITT estimators.
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 - Why biased? $\mathbb{E}[A/B] \neq \mathbb{E}[A] / \mathbb{E}[B]$ when B is random.
 - Why consistent? $B \xrightarrow{P} \mathbb{E}[B]$ (constant) $\rightsquigarrow$ randomness vanishes.
- Equivalent to the **two-stage least squares** estimator:
 - Regress D_i on Z_i to get fitted values, $\widehat{D}_i$.
 - Regress Y_i on $\widehat{D}_i$.

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- Equivalent to the **two-stage least squares** estimator:
 - Regress D_i on Z_i to get fitted values, $\widehat{D}_i$.
 - Regress Y_i on $\widehat{D}_i$.
- We can use the delta method to find the (superpopulation) variance:

$$\mathbb{V}[\widehat{\tau}_{iv}] = \frac{1}{ITT_D^2} \mathbb{V}[\widehat{ITT}_Y] + \frac{ITT_Y^2}{ITT_D^4} \mathbb{V}[\widehat{ITT}_D] - 2 \cdot \frac{ITT_Y}{ITT_D^3} \cdot \text{cov}[\widehat{ITT}_Y, \widehat{ITT}_D]$$

Numerical Example: God's-Eye View

Unit	$D_i(0)$	$D_i(1)$	Type	$Y_i(0,0)$	$Y_i(1, D_i(1))$	τ_i
1	0	1	Complier	4	7	3
2	0	1	Complier	3	7	4
3	0	0	Noncompl.	5	5	–
4	0	1	Complier	2	7	5
5	0	0	Noncompl.	6	6	–
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- One-sided: $D_i(0) = 0$ for all. $\pi_{co} = 5/8, \pi_{nc} = 3/8$.
- **Noncompliers:** Z has no effect on D or Y . τ_i not defined via D .
- **Compliers:** treatment effect is *only* defined for these units.
 - $\tau_{LATE} = (3 + 4 + 5 + 4 + 4)/5 = 4.0$

Numerical Example: What We Actually Observe

Suppose Z is randomly assigned: units 1–4 get $Z = 1$, units 5–8 get $Z = 0$.

Unit	Z_i	D_i	Y_i	Type	(Hidden)
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- $\widehat{\text{ITT}}_Y = \bar{Y}_1 - \bar{Y}_0 = 6.5 - 3.5 = 3.0$
- $\widehat{\text{ITT}}_D = \bar{D}_1 - \bar{D}_0 = 0.75 - 0 = 0.75$
- $\widehat{\tau}_{\text{iv}} = 3.0/0.75 = 4.0 = \tau_{\text{LATE}}$
 - Compliance type is **never observed**, but Wald recovers the LATE!
 - Exact match here is specific to this example. With small n , Wald generally $\neq$ LATE; it converges asymptotically.

4/ Two-Sided Noncompliance

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 - Some may take treatment even without encouragement.
- Z_i is the encouragement, and D_i is the treatment.
- No change in estimation, just different identification assumptions.

Compliance Types

- Four compliance types (or **principal strata**) in this setting:
 - Complier: $D_i(1) = 1$ and $D_i(0) = 0$
 - Always-taker: $D_i(1) = D_i(0) = 1$
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- Connections between observed data and compliance types:

	$Z_i = 0$	$Z_i = 1$
$D_i = 0$	Never-taker or Complier	Never-taker or Defier
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- Let π_{co} , π_{at} , π_{nt} , and π_{df} be the proportions of each compliance type.
- ITT effects on D_i are more murky: $ITT_D = \pi_{co} - \pi_{df}$
 - Defiers really makes things messy!

Instrumental Variables Assumptions

- Canonical IV assumptions for Z_i to be a valid instrument:
 1. Randomization of Z_i
 2. Presence of some compliers $\pi_{co} \neq 0$ (first-stage)
 3. Exclusion restriction $Y_i(z, d) = Y_i(z', d)$
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- $\rightsquigarrow$ Same identification result: $\tau_{LATE} = ITT_Y / ITT_D$

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- Without further assumptions, $\tau_{\text{LATE}} \neq \tau$.
- Complier group depends on the instrument $\rightsquigarrow$ different IVs will lead to different identified estimands.
 - But, we cannot do any better in terms of point estimation without more assumptions.

IV and Mechanisms

- Notice: in the IV setup, D is an **intermediate outcome** between Z and Y .
 - Same DAG as mediation analysis: $Z \xrightarrow{\beta} D \xrightarrow{\delta} Y$, direct: $Z \xrightarrow{\gamma} Y$.

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- Takeaway: IV tells us *whether* D affects Y , not *how*.

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 - 60% are compliers: they open the door if canvassed ($\pi_{co} = 0.6$).
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 - One-sided noncompliance: $D_i(0) = 0$ for all.

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 - People who open the door $\neq$ random sample of the population.
 - This is exactly why as-treated analysis fails!
- True LATE = 5 percentage points (effect of hearing the message on voter turnout, among compliers).

Simulating the DGP

The code generates data consistent with the GOTV setup:

```
1  set.seed(02138); n <- 500
2  complier <- rbinom(n, 1, 0.6)
3  Z <- rbinom(n, 1, 0.5)
4  D <- Z * complier # one-sided noncompliance
5
6  # Compliers have higher baseline turnout (+3 pp)
7  Y0 <- 10 + 3 * complier + rnorm(n)
8  Y1 <- Y0 + 5 * complier # true LATE = 5
9  Y <- D * Y1 + (1 - D) * Y0
```

- $Y_i(0)$: baseline turnout. Compliers average 13, noncompliers 10.
- $Y_i(1) - Y_i(0) = 5$ for compliers, 0 for noncompliers.
- After assignment, we observe only (Z_i, D_i, Y_i) , not compliance type.

Three Estimators, Three Answers

With the simulated data, we can compute and compare all three approaches:

```
1 ITT_Y <- mean(Y[Z==1]) - mean(Y[Z==0]) # 2.96
2 ITT_D <- mean(D[Z==1]) - mean(D[Z==0]) # 0.60
3 Wald <- ITT_Y / ITT_D # 4.91
4 as_tr <- mean(Y[D==1]) - mean(Y[D==0]) # 7.82
```

Estimator	Estimate	Truth	Interpretation
$\widehat{ITT}_Y$	2.96	3.0	Effect of <i>assignment</i> on turnout
$\widehat{ITT}_D (= \hat{\pi}_{co})$	0.60	0.6	Proportion of compliers
$\hat{\tau}_{iv}$ (Wald)	4.91	5.0	Effect of <i>canvassing</i> on compliers
As-treated	7.82	–	Confounded! (+3 from selection)

- As-treated is ≈ 8 instead of 5: the +3 baseline gap inflates the estimate.

Confirming with ivreg

The AER package implements 2SLS. With binary Z and D , this is exactly the Wald estimator:

```
1 library(AER)
2 iv_model <- ivreg(Y ~ D | Z, data = my_data)
3 ols_model <- lm(Y ~ D, data = my_data)
```

	IV (ivreg)	OLS (lm)
Coefficient on D	4.91	7.82
Interpretation	$\hat{\tau}_{\text{LATE}}$	As-treated (biased)

- `ivreg(Y ~ D | Z)`: instrument Z for endogenous D . Consistent for LATE.
- `lm(Y ~ D)`: treats D as exogenous. Biased when D is confounded.
- Because we know the DGP, we can verify: IV recovers the truth, OLS does not.

Pseudo-Code: Wald Estimator

```
1  # *Recall what we did in Neyman's approach*
2  my_data # data includes Z, D, and Y
3
4  # Proportion of compliers (using ITT_D)
5  pi_co <- mean(my_data$D[my_data$Z == 1]) - mean(my_data$D[my_data$Z == 0])
6
7  # Compute ITT's
8  ITT_Y <- mean(my_data$Y[my_data$Z == 1]) - mean(my_data$Y[my_data$Z == 0])
9  ITT_D <- mean(my_data$D[my_data$Z == 1]) - mean(my_data$D[my_data$Z == 0])
10 # (ITT_D = pi_co)
11
12 # Wald estimator
13 Wald_est <- ITT_Y / ITT_D
14
15 # Variance (delta method)
16 Var_ITT_Y <- var(my_data$Y[my_data$Z==1])/sum(my_data$Z==1) +
17             var(my_data$Y[my_data$Z==0])/sum(my_data$Z==0)
18 Var_ITT_D <- var(my_data$D[my_data$Z==1])/sum(my_data$Z==1) +
19             var(my_data$D[my_data$Z==0])/sum(my_data$Z==0)
20
21 # Covariance term (within treated assignment group)
22 demeaned_y <- my_data$Y[my_data$Z == 1] - mean(my_data$Y[my_data$Z == 1])
23 demeaned_d <- my_data$D[my_data$Z == 1] - mean(my_data$D[my_data$Z == 1])
24 denom <- sum(my_data$Z == 1) * (sum(my_data$Z == 1) - 1)
25 Covar_est <- sum(demeaned_y * demeaned_d) / denom
26
27 # Delta method variance
28 Var_Wald <- (1/ITT_D^2)*Var_ITT_Y + (ITT_Y^2/ITT_D^4)*Var_ITT_D -
29             2*(ITT_Y/ITT_D^3)*Covar_est
30
31 # Or use TSLS with AER::ivreg
32 ivmodel <- AER::ivreg(Y ~ D | Z, data = my_data)
33 ivpack::robust.se(ivmodel)
```


5/ In-Class Exercise

In-Class Exercise: Noncompliance in Practice

Setting: A pharmaceutical company runs a randomized trial for a new drug. $n = 200$ patients are randomized: 100 assigned to treatment ($Z_i = 1$), 100 to placebo ($Z_i = 0$). However, some patients in the treatment group refuse to take the drug.

	n	Took drug ($\bar{D}$)	Health score ($\bar{Y}$)
Assigned treatment ($Z = 1$)	100	0.70	75
Assigned placebo ($Z = 0$)	100	0.00	70

Exercise Questions

1. Is this one-sided or two-sided noncompliance? Why?
2. Compute $\widehat{ITT}_Y$, $\widehat{ITT}_D$, and $\widehat{\tau}_{iv}$. Interpret each in words.
3. What is $\widehat{\pi}_{co}$? What proportion of the treatment group are noncompliers?
4. A colleague suggests: “Just compare patients who actually took the drug vs. those who didn’t.” What is wrong with this?
5. Think of a plausible violation of the exclusion restriction in this setting. Is there a channel from Z to Y that bypasses D ?

Enjoy the rest of your weekend! :)

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Appendix

Who Are the Compliers? (Abadie, 2003)

- We cannot identify *which* units are compliers from observed data.
 - Compliance type depends on $(D_i(0), D_i(1))$, both never observed together.
- But we can learn about compliers' **characteristics**.

- For any pretreatment covariate X_i :

$$\mathbb{E}[X_i \mid \text{complier}] = \frac{\mathbb{E}[X_i \cdot D_i \mid Z_i = 1] - \mathbb{E}[X_i \cdot D_i \mid Z_i = 0]}{\mathbb{E}[D_i \mid Z_i = 1] - \mathbb{E}[D_i \mid Z_i = 0]}$$

- Same Wald ratio structure, but with $X_i \cdot D_i$ in place of Y_i .
- Why does this matter?
 - **External validity**: are compliers representative of the population of interest?
 - **Policy relevance**: if a program expands, will new participants look like current compliers?
 - Compare complier means to population means to assess.